

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

On

**Comparative Analysis of Deep Learning and Machine Learning for
Email Spam Detection**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **Comparative Analysis of Deep Learning and Machine Learning for Email Spam Detection** partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Monika Aggarwal**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

Email spam detection has become a critical challenge in modern digital communication due to the rapid growth of unsolicited and malicious messages. Spam emails not only create inconvenience for users but also pose serious security threats, including phishing attacks, malware distribution, and financial fraud. This research paper presents a comparative analysis of machine learning and deep learning approaches for effective email spam detection. The study primarily focuses on the implementation of the Naïve Bayes classifier as a baseline model, utilizing a dataset of labeled email messages categorized as spam or ham.

The proposed methodology involves data preprocessing, feature extraction, and classification using probabilistic techniques based on Bayes' theorem. Experimental results demonstrate that the Naïve Bayes model achieves high accuracy and strong performance in distinguishing between spam and legitimate emails, with evaluation metrics such as precision, recall, F1-score, and ROC-AUC indicating reliable classification capability. Additionally, the study reviews various existing approaches, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and hybrid models like CNN-LSTM, highlighting their improved performance in handling complex patterns in email data.

The findings suggest that while traditional machine learning models like Naïve Bayes are efficient and computationally less expensive, deep learning models offer superior accuracy and adaptability to evolving spam patterns. This research emphasizes the importance of selecting appropriate models and features for building robust spam detection systems. Future work may involve integrating hybrid and deep learning techniques to further enhance detection accuracy and system scalability.

Introduction to Project

Machine learning models have been found effective in many application areas now a days [1]. In the world technology is everywhere, the sharing of information has become very easy as well as fast. There are many platforms where users can share that data very easily [2]. Among the various information sharing platforms, Emails are the simplest, Cheapest and the fastest medium of sharing of information worldwide. But if there is something good than there is something bad also in it, so due to the simplicity of emails, it can be used for the different kind of attacks, and the most efficient one which is currently taking place is as SPAM. As we all know that time is more important than anything, so no body want to receive the unwanted mails which is not important to them. SPAM emails also have malicious contents which is in the form of attachment or any URLs, which is very harmful if we once click on those links, it can host the security breach of the system. Nowadays, commercial or advertisement bulk emails has also become a huge problem for the users. The threat of fake emails is increasing day by day on early basis every year, which is liable for over seventy-seven percentage of the whole worldwide email traffic. This has too led to the significant commercial losses for numerous customers who have come in fraud in cases like internet scams and other fraudulent practices. Fraudsters often send emails posing as reputable companies, attempting to deceive individuals into revealing confidential private details such as passcodes, Financial ID Codes, Numbers (BVN), and Payment card details.

Literature Survey

In the paper titled ‘Efficient Spam Email Classification using Machine Learning Algorithms’, written by Pallavi N & Jaya Rekha [3] the authors focus on the classification of emails into spam by utilizing some measures of machine learning algorithms like Naive, SVMS, Decision Trees & etc. The paper also offers a comprehensive discussion on uses of ML (machine learning) in email fraud classification & how effective these methods are. Nevertheless, a potential concern raised is the concentration on a single class of algorithms what can raise the question whether the results are generalizable to other contexts or other data collections. In the paper “Spam Detection System Using Supervised ML” published by Abhila et al. [4], the paper presents a method for the identification of spam messages from ham messages in an inbox using the Naïve Bayes, a data mining method. The system is well defined with step-bystep sessions for data gathering and

preparation and feature extraction and training and testing for the message classification. It is apparent that Naive Bayes classifiers are very simple and can be implemented very easily for use, but there may be some problems with model complexity, features handling, etc., It is not very adaptable to new spamming techniques. In the work titled “Mansoor et al. [5], the authors summarize about the participation of ML algorithms to enhancing the performance of email spam classification. It covers the problems with conventional methods of entertainmentbased, rule-based filtering and what machine learning techniques can address these issues. The paper, therefore, underscore the need to come up with better normalized systems of detecting spam mails than what we currently have in the market for combating this continuous evolution of the stance of spammers. The authors recommend more study in this field with a view of improving the efficiency of the spam filtering systems. Obviously in his article “ML Approaches to Detect Email Spam Anomaly”, Narendra Kumar et al. [6], the author explains and underlines the significance of using enhanced ML algorithms in detecting heaps of fraud emails. The approach used includes data cleaning where duplicate and punctuation are removed from the data set. NB (Naïve Byes) has been used in this model because of the effectiveness and especially for predictor matching using raw string or regular expression. Another important advantage of the model discussed in this paper over the existing ones is the focus on the inspection of the material of the collected texts in order to conclude possible spam. Also, it overcomes the problem of evaluating emails with small datasets, through implementing different ML for proper finding of fraud mails. A. Karim et al. [7] discuss different security threats relative to emails, mainly fraud emails and potentiality for fraud percentage in their study on smart junk email recognition. They examine at length both the ML and non-machine education techniques of spam identification & filtering. Most of the authors pointed out that several studies of the email spam detection use supervised learning components; they attributed this to the reliability and repeatability of these methods. They also talk about several algorithms and decide that frameworks containing several algorithms are overall better to use than the single one. Further, they observe that most of the existing works in the context of monitoring spam, in particular, phishing, is dependent upon word-based classification or clustering techniques. S. C. Lanka et al. in their work recognized as “Spambased Email Identification and Detection using Machine Learning Techniques” [8] discussed briefly on variety of different ML models common for fraud detection works. Moreover, point out that Naïve Bayes algorithm takes the position of one of the finest efficient procedures for detecting junk mail in spite of its simplicity and high efficiency,

which make this technique preferable as compared with other ML models in the framework of fraud detection. In this paper with the title “Classification and Prediction of Spam Emails Based on AI enabling Models using Deep and Machine Learning Techniques”, J. M. Muhammad et al. [9] present and discuss an approach Rooted on ML and Deep Neural-networking, including Gaussian-Naïve-Bayes (GNB), Convolutional-NeuralNetworks (CNN), Long Short-Term Memory (LSTM) and a compounded model CNN-LSTM. The study as highlighted the above focus of using the open-source of the Spam Assassin dataset with real email samples of approximately 6000 classes for spam categorization prediction. The models were trained and tested and from this experiment it was seen that in this paper CNNLSTM model yielded the perfect results with predictions precision of 98.68% on spam dataset. Another paper with Temporal Naïve Bayes Classifier written by J. Mythili et al. [10], it presents a framework which incorporates temporal features include email arrival time, the sender’s engagement pattern & keyword allocation in this spam detection process. The ability to avoid becoming obsolete with regard to strategies used by spam developers is enhanced by this technique. Performance studies reveal that the Temporal Naive Bayes classifier is more effective than other static classifiers for filtering out junk email and authentic messages going to the spam folder. The findings give emphasis on temporal features when designing better and more resilient email security mechanisms. In the context of literature, the study named ‘Content Based Email Spam Filtering and Detection using Hybrid Supervised Learning Approach’, was conducted by N. U. Saquib et al. [11], described a novel hybrid architecture for the content-based filtering of spam email. The approach employs the feature selection, which uses Word2Vec (W2V) algorithm and bio-inspired Particle Swarm Optimization (PSO), SVM and CNN. The proposed model is implemented with an open-source software, and the accuracy of the model is tested by constructing a confusion matrix. In a current experiment, it is revealed that this hybrid scheme has a better performance of detecting spam emails than previous models. In the paper written by L. Zou et al establish a comparative study between two different types of NLP models for spam email detection. The study compares the count-based model namely the TFIDF-SVD-RF model with the inference-based model namely Word2VecBiGRU-NN model using a real-world email data set. The count-based approach had an accuracy of 99.10% and an F1 score of 96.58%, on the other side, the inferencebased approach had a score rate of 98.39% and an F1 score of 93.92%. The work also indicates the advantages of one approach over the other as well as show that while developing an approach to the spam detection and other text categorization

techniques it is essential to select the appropriate template for better accuracy. The article by A. Saini et al. [13] has enumerated various classification techniques to develop a model to detect spam emails and tried to address the issues arises due to spam emails along with adding their valuable input towards the progress of privacy and security. For the classification methods, the model adopts Naive Bayes, K*, J48 and Random Forest approaches. Of these, the Random Forest model arrived at the highest prediction accuracy of 95.48% and so can be regarded as the most effective classifier in the study. The paper titled by S. T. Singh et al. [14] explores the application of machine learning and deep learning techniques for email spam detection. The study involves the implementation of four algorithms—KNN, Naive Bayes, BiLSTM, and Deep CNN—demonstrating their effectiveness in identifying spam. Among the tested models, the Deep CNN algorithm outperformed the others, achieving the highest accuracy and F1 score, making it the most reliable approach for spam detection in this research.

Researcher(s)	Model Used	Year	Reference
Pallavi N. & Jaya Rekha	Naive Bayes, SVM, Decision Trees	2023	[3]
Abhila et al.	Naive Bayes	2023	[4]
Mansoor et al.	Machine Learning Algorithms	2021	[5]
Narendra et al.	Naive Bayes	2022	[6]
A. Karim et al.	Supervised Learning Algorithms, Multialgorithm	2019	[7]
Lanka et al.	Naive Bayes	2023	[8]
Muhammad et al.	GNB, CNN, LSTM, CNN-LSTM Hybrid Model	2022	[9]
J. Mythili et al.	Temporal Naive Bayes	2024	[10]

Researcher(s)	Model Used	Year	Reference
Saquib et al.	Word2Vec, PSO, SVM, CNN	2024	[11]
L. Zou et al.	TFIDF-SVD-RF, Word2Vec-BiGRU-NN	2024	[12]
Saini et al.	Naive Bayes, J48, K* and Random Forest	2023	[13]
Singh et al.	KNN, Naive Bayes, BiLSTM, and Deep CNN	2023	[14]

Methodology/ Planning of work

We have used the Naïve byes algorithm which is a supervised ML for our proposed system. Using this we are able to easily identify the HAM and SPAM text which is available in the inbox of email. We have taken dataset collection which contains Spam Messages as well as Ham Messages, and that have been stored in the excel sheet. We also done the pre-processing of that excel sheet. The feature selection technique has been used, and this technique is used when there is large amount of data. The dataset goes through both training as well testing phase. We have taken 4789 messages as a dataset in this proposed model. The ratio of testing and training is 25: 75.

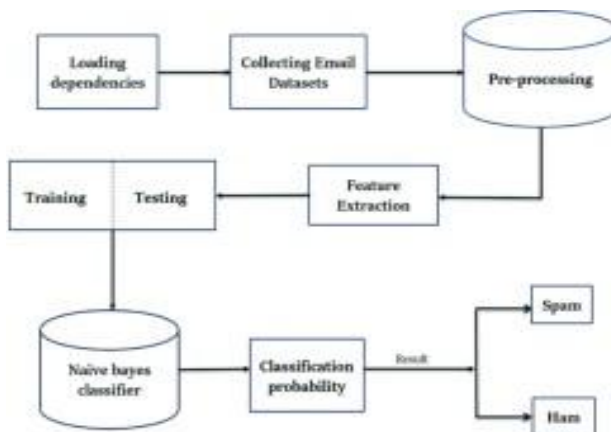


Figure 1: Diagrammatically representation of our model.

Figure [1] illustrates the workflow of a spam detection model using a Naïve Bayes classifier. It begins with loading the necessary dependencies and collecting email datasets, which are then sent through a pre-processing stage to clean and prepare the text for analysis. After preprocessing, the data undergoes feature extraction, converting the text into numerical features that the model can understand. The dataset is then split into training and testing portions, with the training data used to train the Naïve Bayes classifier. Once trained, the model generates classification probabilities that determine whether a new message is spam or ham. The final output categorizes each email based on these probabilities, completing the spam classification process.

Algorithm

Naive Bayesian Classifier is a probabilistic machine learning model designed specifically for text classification; Naive Bayesian is one of the primary methods for spam message detection. It is very efficient and useful for training datasets much faster than other classifiers.

This is a word and phrases pattern which correlate from the spam and ham dataset. Then they calculate some probabilities through Bayes' theorem to predict if a message is spam or ham. Statistical probability techniques are used as a mining method to filter or identify the messages in a very effective manner.

The system uses the multinomial Naïve Bayes method, which focuses on the frequency of the phrases. Giving the classifier the ability to differentiate spam from non-spam by calculating how often a word is found in the data sample.

Term Frequency = Each time the term w appears in data(P) / Total no. of terms In the dataset

We can use the formula of Bayes theorem to calculate the message classification probability. This mathematically calculates the probability of whether the provided input text is spam i mean if it has any of the spam words:

P(Spam word)

$$= p(\text{Spam}) * p(\text{word} | \text{Spam}) / (P(\text{Spam}) * p(\text{word} | \text{Spam}) + p(\text{ham}) * p(\text{word} | \text{ham}))$$

Dataset

index	Category	Message	Spam
0	ham	Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...	0
1	ham	Ok lar... Joking wif u oni...	0
2	spam	Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's	1
3	ham	U dun say so early hor... U c already then say...	0
4	ham	Nah I don't think he goes to usf, he lives around here though	0
5	spam	FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, â€1.50 to rcv	1
6	ham	Even my brother is not like to speak with me. They treat me like aids patient.	0
7	ham	As per your request 'Melle Melle (Oru Minnaminunginte Nuringu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune	0
8	spam	WINNER!! As a valued network customer you have been selected to receivea â€900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.	1
9	spam	Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030	1

The image shows a preview of a dataset containing text messages along with their labels indicating whether each message is "ham" (legitimate) or "spam." The table includes columns for the message index, the category label, the message content, and a binary "Spam" column where 0 represents ham and 1 represents spam. The sample entries show regular conversational messages labeled as ham, while promotional or scam-like messages—such as prize notifications, competitions, or urgent marketing texts—are labeled as spam. This dataset is typically used for training and evaluating spam detection models in natural language processing.

Result and Discussion

We have used Naïve Bayesian Classifier model for identifying spam and ham messages. The prediction whether it is spam or not is based on individual calculation of frequency of words which is have been described in this paper, moreover classified on the prediction founded on Theorem of NB Classifier making each phase followed in the process to give a prediction. The result says the categorization to which the message is either spam or ham.

Figure 2, defining the percentage of Ham and spam data respectively, figure 3, visualization of frequent spam words.

Distribution of Spam vs Ham

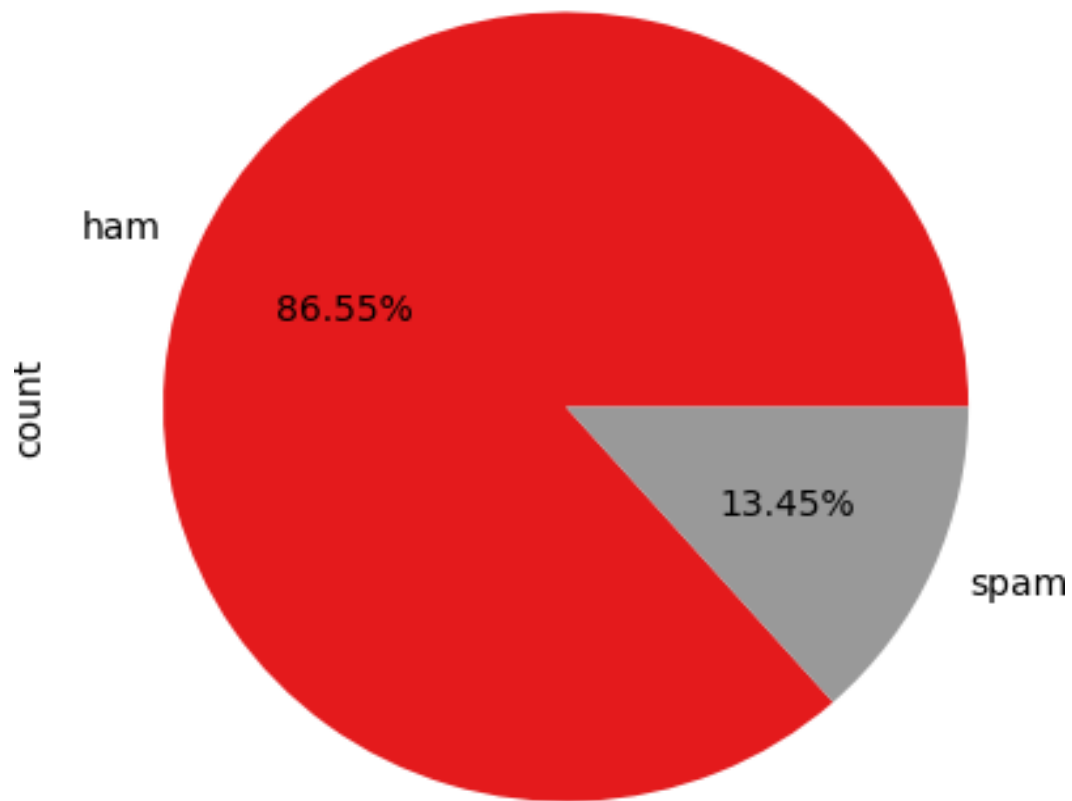


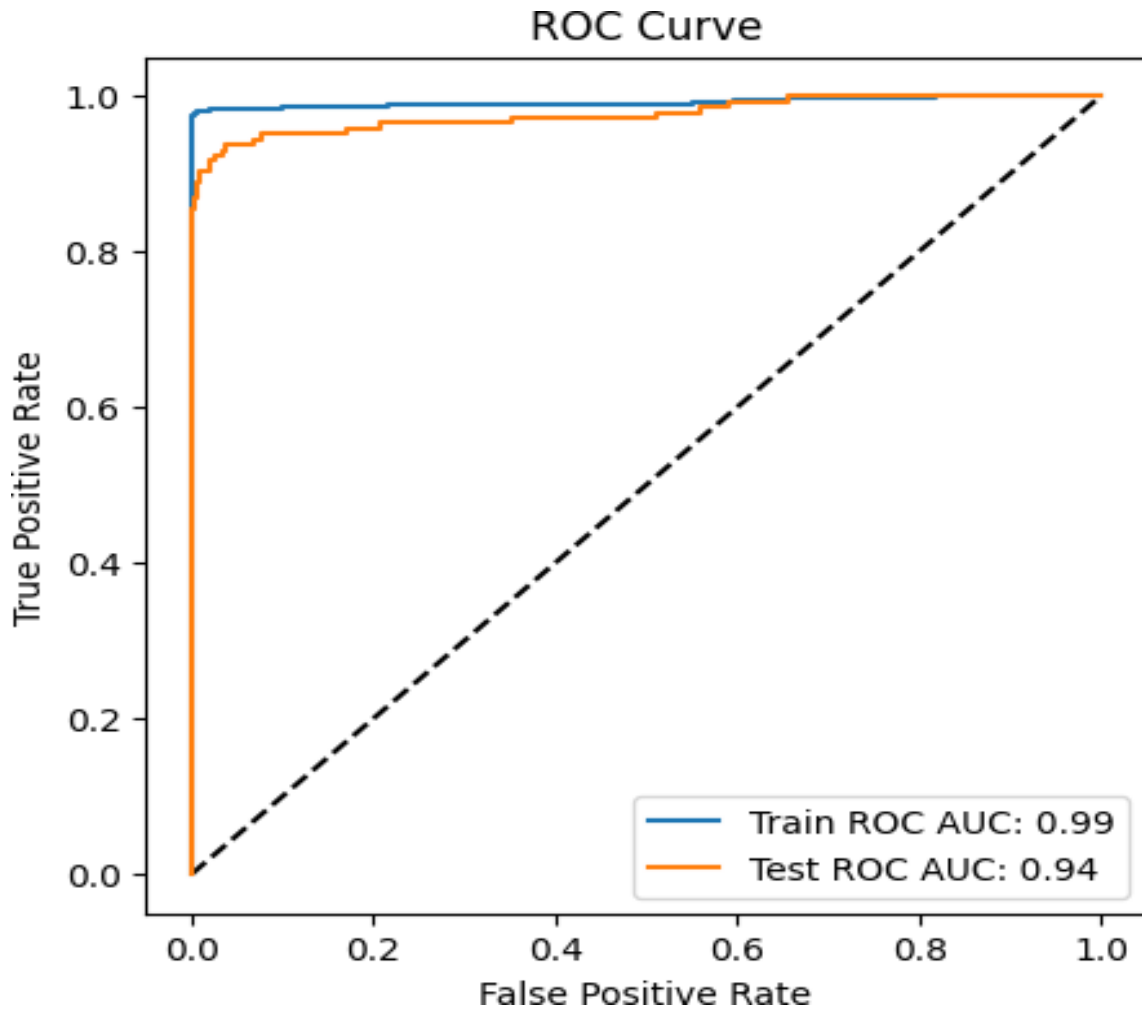
Figure 2: Visualization of output

The pie chart illustrates the distribution of **ham (non-spam)** and **spam** messages within the dataset. It shows that **ham messages make up the majority**, accounting for **86.55%**, while **spam messages represent only 13.45%** of the total. This imbalance indicates that the dataset is dominated by legitimate messages, which is common in real-world communication data. Such class imbalance is important to consider when building spam detection models, as it may influence model performance and require techniques like resampling or class weighting to ensure accurate spam classification.

[illegible]

Figure 3: Spam visualization of frequent spam words

The image is a word cloud titled **“Most Used Words in Spam Messages,”** visually representing the most frequent terms found in spam texts. Larger and darker words, such as **“free,” “call,” “text,” “mobile,” “reply,” “won,”** and **“urgent,”** appear more often in spam messages and therefore dominate the image. Smaller words like **“claim,” “receive,” “club,” “tone,” “service,”** and many others also appear, indicating their recurring presence in promotional or fraudulent messages. The varying font sizes and colors help highlight the relative importance of each term, making it clear which words are most commonly used by spammers to attract attention, create urgency, or deceive recipients.



The image shows a ROC Curve that evaluates the performance of a binary classification model by plotting the True Positive Rate against the False Positive Rate at various threshold settings. The blue curve represents the model's performance on the training data with a very high AUC of 0.99, indicating near-perfect classification, while the orange curve shows the test performance with an AUC of 0.94, reflecting strong generalization to unseen data. Both curves lie well above the diagonal dashed line, which represents random guessing, demonstrating that the model effectively distinguishes between the two classes. The slight drop from train to test AUC suggests minimal overfitting, and overall the model exhibits excellent discriminative ability.

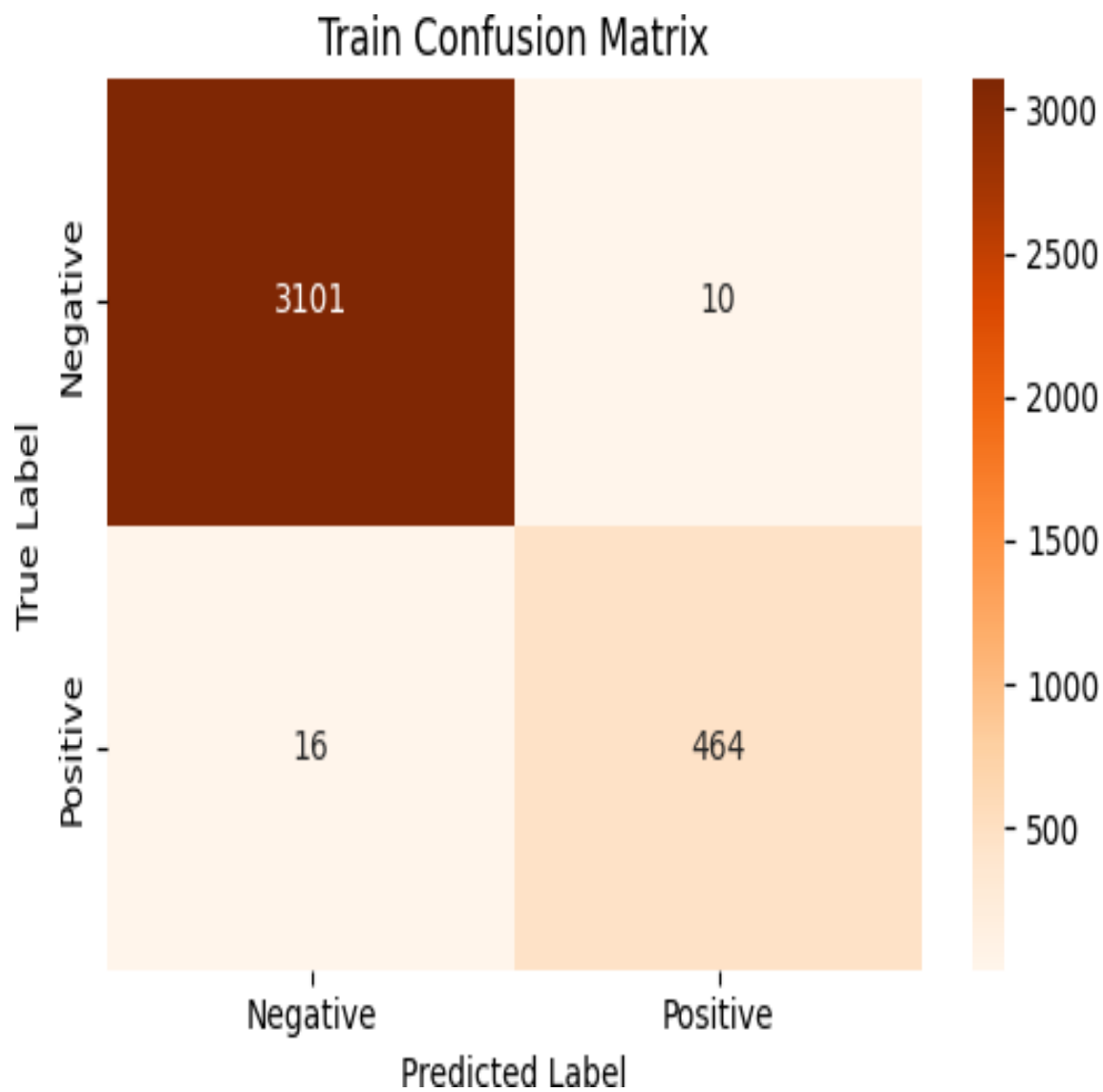


Figure 5: Train Confusion Matrix

Figure [5] defines that, The Train Confusion Matrix shows that the algorithm has been performed well on the training data, with 3101 true negatives and 464 true positives, indicating correct classifications. It has minimal misclassifications, with only 10 false positives that is FP and 16 false negatives that is FN, reflecting high accuracy and good model performance.

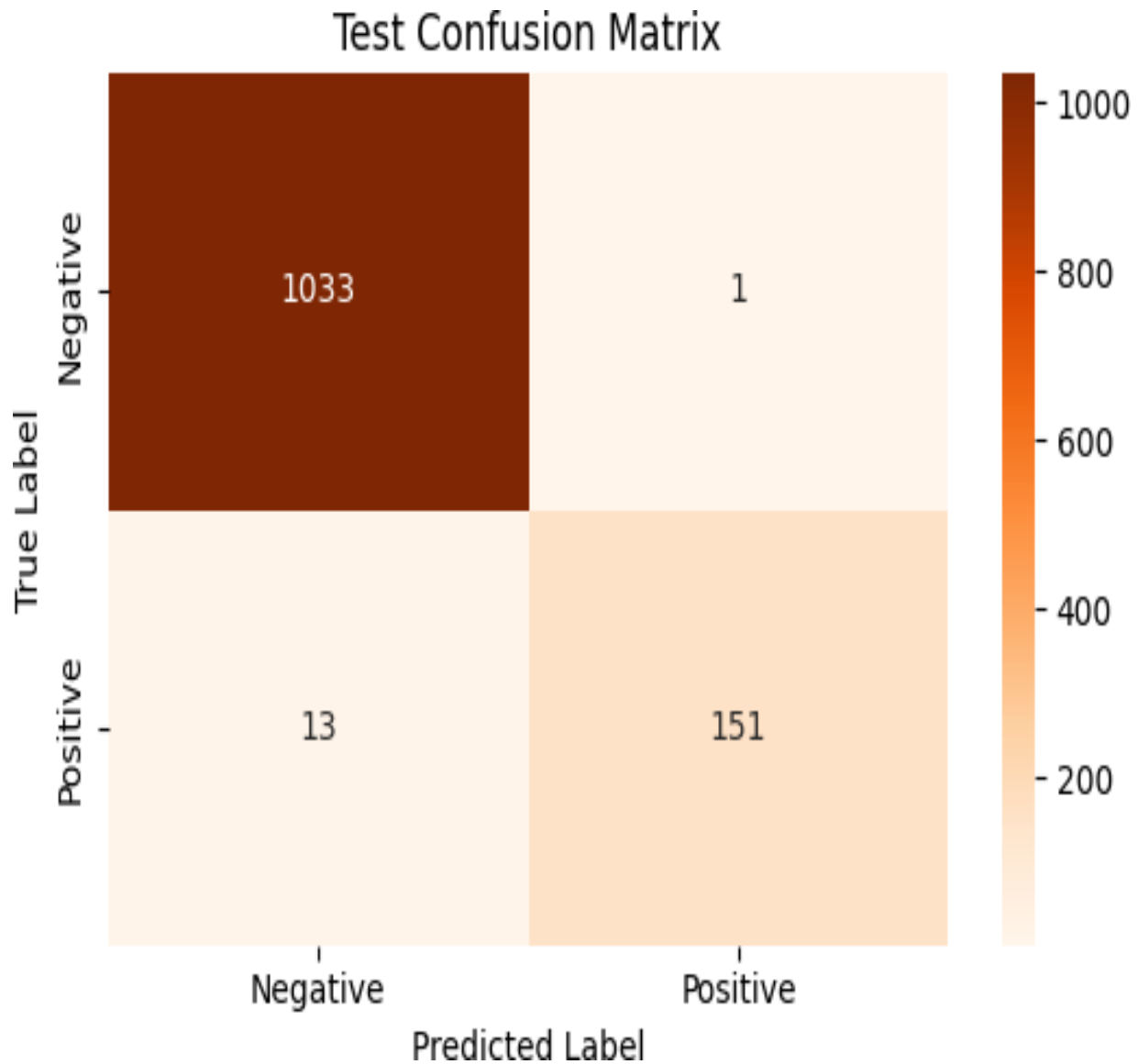


Figure 6: Test confusion matrix

Figure [6] defines that the confusion matrix illustrates the performance of the classification model on the test dataset, showing 1033 true negatives and 151 true positives, indicating the correct classifications of negative and positive instances, respectively. The model has minimal misclassifications, with only 1 false positive (negative instance classified as positive) and 13 false negatives (positive instances classified as negative). This highlights the model's high accuracy and reliability, effectively distinguishing between the two classes.

Train Classification Report:

	precision	recall	f1-score	support
:-----	-----	-----	-----	-----
0	0.996127	0.998382	0.997253	3091
1	0.989858	0.976	0.98288	500
accuracy	0.995266	0.995266	0.995266	0.995266
macro avg	0.992992	0.987191	0.990067	3591
weighted avg	0.995254	0.995266	0.995252	3591

Figure 7: Train Classification Report

Figure [7] shows the **training classification report** for a spam detection model, summarizing its performance across key metrics. Class **0 (ham)** achieves exceptionally high precision (0.9961), recall (0.9984), and F1-score (0.9973), indicating that the model identifies legitimate messages with near-perfect accuracy. Class **1 (spam)** also performs strongly with a precision of 0.9899, recall of 0.9760, and F1-score of 0.9829, showing the model is highly effective at detecting spam, though it misses a small portion of spam messages. The overall accuracy is **99.53%**, reflecting excellent performance, while the macro and weighted averages further confirm that the classifier maintains balanced and reliable results across both classes.

Test Classification Report:

	precision	recall	f1-score	support
:-----	-----	-----	-----	-----
0	0.984948	0.993359	0.989136	1054
1	0.948148	0.888889	0.917563	144
accuracy	0.980801	0.980801	0.980801	0.980801
macro avg	0.966548	0.941124	0.953349	1198
weighted avg	0.980525	0.980801	0.980532	1198

Figure 8: Test Classification Report

Figure [8] shows the test classification report for the spam detection model, summarizing how well it performs on unseen data. Class 0 (ham) has very high precision (0.9849), recall (0.9934), and an F1-score of 0.9891, meaning the model correctly identifies most legitimate messages with minimal mistakes. Class 1 (spam) performs slightly lower but still strong, with a precision of 0.9481, recall of 0.8889, and F1-score of 0.9176, indicating the model occasionally misses some spam but is still highly effective overall. The model's accuracy is 98.08%, showing excellent overall performance on the test set. The macro and weighted averages further confirm that the classifier maintains balanced performance across classes, though the drop in recall for spam compared to training suggests mild overfitting but strong generalization.

Facilities required for proposed work

Software Requirements

1. Programming Language:

- **Python** - Python is a versatile, high-level programming language widely used in Machine Learning (ML) due to its simplicity, readability, and extensive ecosystem of libraries and frameworks. Its syntax is intuitive, making it easier for developers and data scientists to prototype, experiment, and implement complex ML algorithms without getting bogged down by low-level details. Python supports a rich set of libraries like NumPy and Pandas for data manipulation, Matplotlib and Seaborn for visualization, and Scikit-learn, TensorFlow, and PyTorch for building and training machine learning models. Its active community continuously contributes tools, tutorials, and pre-built models, which accelerates development and experimentation. Additionally, Python integrates well with other technologies, making it suitable for deploying ML solutions in real-world applications. Its combination of simplicity, flexibility, and a robust ecosystem makes Python the preferred language for most ML practitioners.

2. Development Environment / IDE:

- **Google Colab** - Google Colab is a free, cloud-based platform provided by Google that allows users to write and run Python code in an online Jupyter Notebook

environment. It is widely used for machine learning, data analysis, and deep learning because it offers free access to powerful hardware like GPUs and TPUs. Colab requires no installation, supports easy sharing and collaboration, and integrates seamlessly with Google Drive, making it a convenient tool for learning, experimenting, and developing AI projects.

3. Libraries and Frameworks:

- **NumPy** – NumPy is a fundamental Python library used for numerical computing, providing support for working with large, multi-dimensional arrays and matrices along with a wide collection of high-performance mathematical functions. It forms the backbone of many data science and machine learning workflows due to its speed, efficiency, and ability to perform vectorized operations, which allow calculations to be executed much faster than with traditional Python lists.
- **Pandas** – Pandas is a powerful Python library used for data manipulation and analysis, offering easy-to-use data structures like **Series** and **DataFrame** for handling structured data. It provides tools for cleaning, transforming, filtering, and aggregating datasets, making it essential for tasks such as data preprocessing, exploratory data analysis, and working with tabular data efficiently.
- **Matplotlib** – Matplotlib is a widely used Python library for creating static, animated, and interactive visualizations. It provides extensive tools for generating a variety of plots—such as line charts, bar graphs, pie charts, scatter plots, and histograms—making it essential for data analysis and scientific visualization. Its flexibility and ability to customize nearly every aspect of a plot make it a foundational tool for presenting data clearly and effectively.
- **Scikit-learn** – Scikit-learn is a popular and powerful Python library used for machine learning, providing simple and efficient tools for data mining and data analysis. It includes a wide range of algorithms for classification, regression, clustering, dimensionality reduction, model selection, and preprocessing. Built on top of NumPy, SciPy, and Matplotlib, Scikit-learn offers a clean and consistent interface that makes it easy to build,

train, evaluate, and deploy machine learning models, making it one of the most widely used frameworks in the data science community.

- **XGBoost** – XGBoost (Extreme Gradient Boosting) is a highly efficient and scalable machine learning library designed for fast and accurate gradient-boosted decision trees. It is widely used in competitive machine learning because it handles large datasets, missing values, and complex patterns exceptionally well. XGBoost includes advanced features such as regularization to prevent overfitting, parallelized tree construction, and built-in handling of imbalanced data, making it one of the top-performing algorithms for structured or tabular data in classification and regression tasks.
- **TensorFlow** – TensorFlow is an open-source deep learning framework developed by Google that provides a flexible platform for building, training, and deploying machine learning models—especially neural networks. It supports a wide range of tasks, including image recognition, natural language processing, and time-series forecasting, and offers high performance through GPU and TPU acceleration. With its computational graph structure, extensive ecosystem (such as Keras for high-level model building), and deployment options for mobile, web, and production environments, TensorFlow is one of the most widely used tools for developing advanced AI systems.

4. Database :

- **Kaggle.com** - Kaggle.com is a popular online platform for data science and machine learning enthusiasts where users can access datasets, participate in competitions, collaborate on projects, and learn through notebooks and courses. It provides a rich collection of publicly available datasets, cloud-based computational notebooks, and community-driven discussions, making it an excellent environment for practicing data analysis, building models, and improving machine learning skills.

Conclusion

In this research, an email spam detection system was developed using the Multinomial Naïve Bayes classifier. The model successfully classified emails into spam and ham categories with high accuracy of over 98%, along with strong precision, recall, F1-score, and ROC-AUC performance. The confusion matrix also showed very few false positives and false negatives, making the system reliable and efficient.

The study highlights that while Naïve Bayes is simple, fast, and computationally efficient, advanced deep learning models such as CNN and LSTM can better capture complex text patterns. However, they require larger datasets and higher computational resources.

Overall, the proposed system provides an effective and low-cost solution for email spam detection and can serve as a strong baseline for future improvements using deep learning and advanced NLP techniques.

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